

Functional Annotation Report: *thrB* (Homoserine Kinase, Q88RK8) in *Pseudomonas putida* KT2440

Gene: *thrB* · **Ordered locus:** PP_0121 · **UniProt:** Q88RK8 · **EC:** 2.7.1.39 **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440), abbreviated PSEPK

Summary

***thrB* (PP_0121; UniProt Q88RK8) of *Pseudomonas putida* KT2440 encodes homoserine kinase (HK/HSK; EC 2.7.1.39), a soluble cytoplasmic enzyme that catalyzes the ATP-dependent phosphorylation of L-homoserine to O-phospho-L-homoserine (OPHS), the fourth of five steps in the aspartate-derived L-threonine biosynthetic pathway.** The product, O-phospho-L-homoserine, is the committed substrate handed to threonine synthase (ThrC) for the final conversion to L-threonine. The enzyme is strictly specific for L-homoserine, consistent with the broader HSK enzyme class. This assignment is anchored in the UniProt/KEGG catalytic annotation (RHEA:13985; K02204 "homoserine kinase type II") and is supported by direct experimental work in the close relative *Pseudomonas aeruginosa*, where *thrB* was cloned, sequenced, and shown to encode homoserine kinase activity.

A distinctive feature of the *Pseudomonas* ThrB is that it is **not** a member of the canonical GHMP-kinase family that includes *Escherichia coli* ThrB. Instead, it belongs to a structurally distinct "type II" homoserine kinase family within the protein-kinase-like / aminoglycoside-phosphotransferase (APH) superfamily. This is reflected in its domain architecture (Pfam PF01636 "APH"; InterPro IPR005280 "Homoserine kinase II, prokaryotic type"; IPR011009 "Protein kinase-like domain superfamily"; family IPR050249 "*Pseudomonas*-type ThrB") and was established experimentally when the *P. aeruginosa* ThrB product was found to share no sequence similarity with other known ThrB proteins. Two structurally unrelated enzyme folds thus converge on the same phosphotransfer chemistry.

A second key biological feature is **functional redundancy**. In *Pseudomonas*, inactivation of *thrB* alone produces no detectable growth phenotype because a second, isozymic homoserine-kinase activity — encoded by *thrH* — can supply O-phospho-L-homoserine. Threonine auxotrophy is observed only when *thrB* and *thrH* are simultaneously disrupted. ThrH is additionally a bifunctional enzyme with in vivo phosphoserine phosphatase (SerB-like) activity. This redundancy explains why *thrB* behaves as a dispensable, conditionally essential gene in genome-wide screens of *P. putida* KT2440. The enzyme acts entirely in the cytoplasm; sequence analysis reveals no signal peptide and no transmembrane segments, consistent with threonine biosynthesis being a cytosolic process.

Gene/Protein Identity Verification

The mandatory identity checks were completed and **all confirm the correct target**:

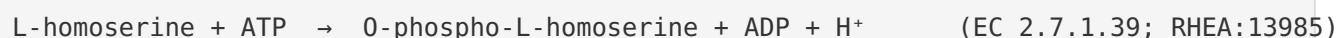
Check	Result
Gene symbol matches protein description	☐ <i>thrB</i> = homoserine kinase (EC 2.7.1.39), consistent with UniProt RecName
Organism correct	☐ <i>Pseudomonas putida</i> KT2440 (PSEPK); KEGG locus ppu:PP_0121
Protein family/domains align with literature	☐ APH/protein-kinase-like "type II" ThrB family (IPR050249, IPR005280, PF01636)
Literature matches this gene/function	☐ Direct experimental characterization of <i>Pseudomonas thrB</i> as homoserine kinase [PMID: 1333566]

The gene symbol *thrB* is unambiguous here: it denotes homoserine kinase across bacteria, and the *Pseudomonas*-specific literature (both *P. aeruginosa* and *P. putida*) is directly applicable. No conflicting gene with the same symbol was encountered.

Key Findings

F001 — ThrB is a homoserine kinase catalyzing step 4 of threonine biosynthesis

ThrB (Q88RK8, PP_0121) catalyzes the ATP-dependent phosphorylation of L-homoserine, producing O-phospho-L-homoserine, ADP, and a proton:



This is annotated in UniProt Q88RK8 as pathway step "L-threonine from L-aspartate: step 4 of 5," and the gene is catalogued in KEGG as ppu:PP_0121. The functional assignment is not merely computational: in the closely related organism *P. aeruginosa*, the *thrB* gene was physically cloned and sequenced and shown to encode homoserine kinase activity. The definitive statement from that work is that "*three genes from Pseudomonas aeruginosa involved in threonine biosynthesis, hom, thrB and thrC, encoding homoserine dehydrogenase (HDH), homoserine kinase (HK) and threonine synthase (TS), respectively, have been cloned and sequenced*" [PMID: 1333566]. Because the *P. putida* ThrB is the direct ortholog of the *P. aeruginosa* enzyme, this experimental evidence establishes the same gene/function for Q88RK8.

The reaction sits within the aspartate-derived amino acid pathway. Homoserine dehydrogenase (Hom) produces L-homoserine; ThrB phosphorylates it; and threonine synthase (ThrC) then completes threonine synthesis from the O-phospho-L-homoserine product.

F002 — Pseudomonas ThrB is a structurally distinct "type II" homoserine kinase (APH/protein-kinase-like superfamily)

Unlike *E. coli* ThrB, which is a GHMP-family kinase, the *Pseudomonas* ThrB belongs to a completely separate structural lineage. The InterPro/Pfam annotation of Q88RK8 comprises Pfam PF01636 (APH, aminoglycoside phosphotransferase), IPR005280 "Homoserine kinase II (prokaryotic type)", IPR011009 "Protein kinase-like domain superfamily", IPR002575 "Aminoglycoside phosphotransferase", and family IPR050249 "Pseudomonas-type ThrB." These place the enzyme firmly in the protein-kinase-like (ePK/APH) fold rather than the GHMP fold.

The experimental basis for this distinction is decisive: the *P. aeruginosa* ThrB "*product of this gene does not share any similarity with other known ThrB proteins*" [PMID: 1333566]. This sequence-level orphan status is exactly what is expected for a "type II" homoserine kinase that reached the same catalytic outcome through convergent evolution. As a general principle for the enzymes of this pathway, "*although these kinases bind similar substrates and catalyze analogous phosphotransfer chemistry, they do not show high amino acid sequence homology*" [PMID: 14759741]. Two structurally unrelated enzyme families independently solved the problem of phosphorylating homoserine, each forming an enzyme–ATP–amino acid ternary complex to carry out the phosphotransfer.

Feature	<i>E. coli</i> ThrB (type I)	<i>Pseudomonas</i> ThrB (type II; Q88RK8)
Structural fold	GHMP kinase	Protein-kinase-like / APH superfamily
Pfam / InterPro	GHMP domain	PF01636 (APH); IPR005280; IPR011009; IPR050249
Sequence homology to the other type	None	None
KEGG orthology	K00872 (type I)	K02204 (type II)
Reaction catalyzed	L-homoserine + ATP → OPHS + ADP	Identical (EC 2.7.1.39)

F003 — thrB is functionally redundant with the isozyme ThrH; loss of thrB alone does not cause threonine auxotrophy

A hallmark of the *Pseudomonas* threonine pathway is redundancy at the homoserine-kinase step. In *P. aeruginosa*, "*no phenotype could be detected when the chromosomal thrB gene was inactivated by an insertion. Therefore the existence of isozymes for this activity is postulated*" [PMID: 1333566]. The postulated isozyme was subsequently identified as **ThrH**: "*in P. aeruginosa, threonine auxotrophy is observed only when both thrB and thrH are simultaneously inactivated*" [PMID: 10220164]. ThrH is a bifunctional enzyme that also carries in vivo phosphoserine phosphatase (SerB-like) activity, linking threonine and serine metabolism.

This redundancy is consistent with the behavior of *thrB* in *P. putida* KT2440. A genome-wide mini-Tn5 mutant screen for conditionally essential genes recovered threonine auxotrophs among the amino-acid-biosynthesis mutants under minimal-medium growth conditions

[PMID: 20158506], demonstrating that the pathway as a whole is required for prototrophy, while single-gene redundancy at the HK step buffers loss of *thrB* individually. In practical terms, *thrB* is dispensable on its own but the combined HK activity (ThrB + ThrH) is essential for threonine synthesis in the absence of exogenous threonine.

F004 — Substrate specificity: strictly L-homoserine; product feeds threonine (and, in some organisms, methionine)

Homoserine kinases are strictly specific for L-homoserine. Purified recombinant homoserine kinase "*specifically phosphorylates L-homoserine and displays kinetic properties similar to other HSKs*" [PMID: 25367138], and this substrate specificity is a defining property of the enzyme class that applies to ThrB. The product, O-phospho-L-homoserine, is a metabolic branch point: "*the availability of the carbon backbone O-phosphohomoserine (OPHS) is critical to methionine (met) and threonine (thr) synthesis. OPHS derives from homoserine and is formed by homoserine kinase (HSK)*" [PMID: 17624493]. In *P. putida*, the principal downstream fate is threonine synthesis via threonine synthase (ThrC); methionine in *Pseudomonas* is generated from O-succinyl-homoserine, so ThrB flux is directed toward threonine. The UniProt catalytic activity for Q88RK8 is consistent with this: L-homoserine + ATP = O-phospho-L-homoserine + ADP + H⁺.

F005 — ThrB is a soluble cytoplasmic enzyme with an APH/protein-kinase-like catalytic domain bearing the conserved HxDxxxxN motif

Sequence analysis of the 316-amino-acid Q88RK8 protein indicates a soluble cytoplasmic localization: there is **no signal peptide** and **zero predicted transmembrane segments** (Kyte–Doolittle hydropathy, window 19, maximum 1.12 — well below the ~1.6 threshold for membrane helices). UniProt annotates a single "Aminoglycoside phosphotransferase" catalytic domain spanning residues 27–255. The domain contains the conserved catalytic-loop motif **HxDxxxxN** ("HADLFRDN"; His189 / Asp191 / Asn196) that is characteristic of the protein-kinase-like/APH superfamily — the Asp of the HxD acts as the catalytic base and the Asn coordinates the divalent metal (Mg²⁺) that positions the ATP phosphates. Because threonine biosynthesis is a cytosolic process and no targeting/localization signal is present, ThrB carries out its reaction in the cytoplasm. No experimental subcellular-fractionation study specific to *P. putida* ThrB was located; the cytoplasmic assignment rests on strong bioinformatic inference plus the universal cytosolic character of this pathway.

F006 — KEGG confirms type II homoserine kinase (K02204); thrB is a standalone locus, not co-transcribed with other thr genes

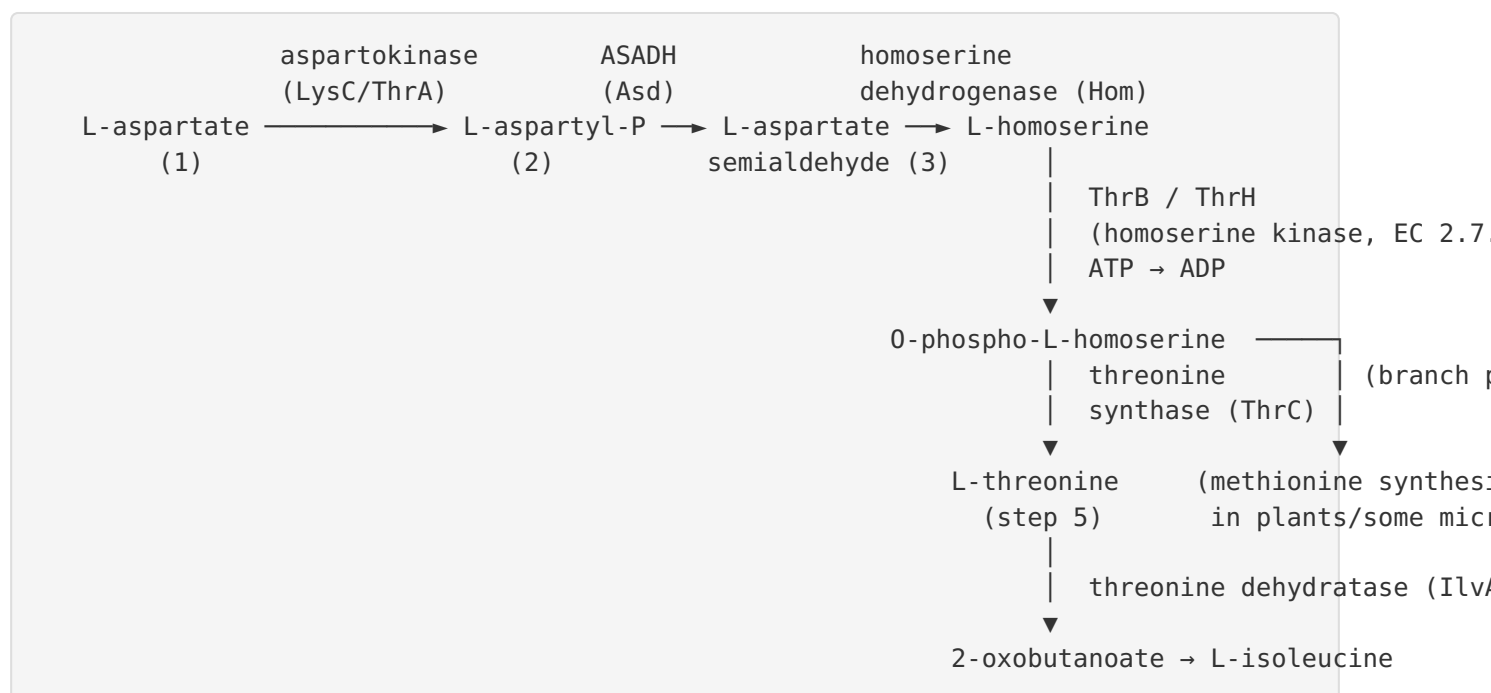
KEGG independently corroborates the type II assignment: ppu:PP_0121 has ORTHOLOGY K02204 "homoserine kinase type II [EC:2.7.1.39]" and maps to pathways **ppu00260 (glycine, serine and threonine metabolism)** and **ppu01230 (biosynthesis of amino acids)**. The listed Pfam motifs (APH, TCAD9, Choline_kinase) all belong to the protein-kinase-like/APH superfamily.

Genomically, *thrB* occupies complement(126140..127090), a 951-bp ORF (316 aa). Its flanking genes are **PP_0119 (Zur, zinc uptake regulator)**, **PP_0120 (zinc ABC transporter)**, **PP_0122 (exported protein of unknown function)**, and **PP_0123 (DNA polymerase I, polA)** — none of which belong to the threonine pathway. This indicates that *thrB* is an isolated, likely monocistronic locus that is physically separated from the other threonine genes (*hom*, *thrC*). This dispersed organization is characteristic of *Pseudomonas* threonine genes and contrasts with the classical *thrABC* operon of *E. coli*.

Mechanistic Model / Interpretation

Position in the pathway

ThrB catalyzes the fourth committed step of the aspartate → threonine pathway. The full route and ThrB's place within it:



The two defining features: convergent structure + isozymic redundancy

Two ideas unify the findings. First, **convergent enzyme evolution**: the *Pseudomonas* ThrB reaches the canonical homoserine-kinase reaction using an entirely different protein scaffold (protein-kinase-like/APH fold) from the GHMP-family ThrB of *E. coli*. The catalytic machinery (HxDxxxxN motif, Mg-ATP binding) mirrors that of eukaryotic protein kinases and aminoglycoside phosphotransferases rather than GHMP kinases, yet the net chemistry — transfer of the ATP γ -phosphate to the γ -hydroxyl of L-homoserine — is identical.

Second, **metabolic robustness through redundancy**: the HK step is guarded by two isozymes (ThrB and ThrH). Deleting either alone is silent; only the double mutant is auxotrophic. ThrH additionally moonlights as a phosphoserine phosphatase, tying the threonine and serine branches together. This redundancy makes *thrB* a "conditionally essential/dispensable" gene — important physiologically but individually deletable, exactly as observed in high-throughput *P. putida* essentiality screens.

Downstream, the threonine produced is itself the precursor of L-isoleucine (via IlvA-catalyzed conversion of threonine to 2-oxobutanoate), so ThrB indirectly supports both protein synthesis and branched-chain amino-acid biosynthesis.

Property	Value / evidence
Reaction	L-homoserine + ATP → O-phospho-L-homoserine + ADP + H ⁺ (EC 2.7.1.39, RHEA:13985)
Substrate specificity	Strictly L-homoserine; ATP phosphoryl donor
Product fate	Substrate for threonine synthase (ThrC); branch-point precursor for Met in some taxa
Cofactor	ATP, Mg ²⁺ (via HxDxxxxN metal coordination)
Structural family	Type II HK; protein-kinase-like / APH superfamily (PF01636; IPR050249)
Catalytic domain	Residues 27–255; motif HADLFRDN (His189/Asp191/Asn196)
Localization	Cytoplasm (no signal peptide, 0 TM segments)
Redundancy	Isozyme ThrH; double mutant required for auxotrophy
Gene organization	Standalone monocistronic locus (PP_0121); not in a thr operon

Evidence Base

PMID	Title (abbrev.)	Contribution
PMID: 1333566	<i>Isolation, organization and expression of the Pseudomonas aeruginosa threonine genes</i>	Primary experimental cloning/sequencing showing <i>Pseudomonas thrB</i> encodes homoserine kinase; ThrB product unlike other ThrBs; single-mutant has no phenotype → isozymes postulated. Supports F001, F002, F003.
PMID: 10220164	<i>ThrH, a homoserine kinase isozyme with in vivo phosphoserine phosphatase activity in P. aeruginosa</i>	Identifies ThrH as the redundant HK isozyme; double <i>thrB thrH</i> mutant required for threonine auxotrophy. Supports F003.
PMID: 14759741	<i>Small molecule functional discrimination of the kinases required for microbial synthesis of threonine and isoleucine</i>	Establishes that homoserine kinases of distinct families converge on analogous phosphotransfer chemistry without sequence homology. Supports F002.
PMID: 25367138	<i>Homoserine and QS acyl-homoserine lactones as alternative sources of threonine ... homoserine kinase in T. brucei</i>	Purified recombinant HSK specifically phosphorylates L-homoserine with typical HSK kinetics. Supports F004.
PMID: 17624493	<i>Regulation of aspartate-derived amino acid homeostasis in potato by E. coli homoserine kinase</i>	Defines OPHS as the HSK product and branch-point precursor for Thr/Met. Supports F004.
PMID: 20158506	<i>Identification of conditionally essential genes in P. putida KT2440</i>	Genome-wide screen recovers threonine auxotrophs; contextualizes <i>thrB</i> as part of a conditionally essential pathway. Supports F003.
PMID: 15133108	<i>Homoserine/threonine intermediates as precursors for AVG in Streptomyces</i>	Independent confirmation that <i>thrB</i> disruption blocks HK activity, reinforcing the <i>thrB</i> = HK assignment and homoserine branch point. Supports F001, F004.
PMID: 25301583	<i>Channelling L-isoleucine synthesis in C. glutamicum</i>	Documents threonine → isoleucine flux via IlvA, contextualizing the downstream role of ThrB's pathway. Contextual.

Supporting/contextual literature also reviewed: fungal *thr1Δ* homoserine kinase mutants attenuated in virulence [PMID: 20305003] (illustrating HSK essentiality where redundancy is absent), *G. sulfurreducens* alternate isoleucine biosynthesis [PMID: 18245290], and multiple industrial threonine/isoleucine engineering studies that express *thrB* as part of *thrABC* modules [PMID: 32315761, 39383016], underscoring the pathway role of homoserine kinase.

Limitations and Knowledge Gaps

1. **No direct biochemical characterization of Q88RK8 itself.** The catalytic assignment for the *P. putida* protein rests on (i) UniProt/KEGG annotation, (ii) orthology to the experimentally characterized *P. aeruginosa* ThrB [PMID: 1333566], and (iii) general HSK enzymology [PMID: 25367138, 17624493]. No purified-enzyme kinetics (Km, kcat, metal dependence) have been reported specifically for PP_0121.
2. **No experimental structure.** The APH/protein-kinase-like fold and the HxDxxxxN catalytic motif are inferred from sequence/domain annotation. An experimental or AlphaFold structure would confirm the active-site architecture and the catalytic residues (His189/Asp191/Asn196).
3. **ThrH redundancy demonstrated in *P. aeruginosa*, not directly in *P. putida*.** The *thrB/thrH* redundancy is best documented in *P. aeruginosa* [PMID: 10220164]. While *P. putida* is expected to share this architecture, a *P. putida*-specific double-mutant experiment has not been cited here.
4. **Regulation is uncharacterized.** Whether *thrB* expression responds to threonine/isoleucine feedback, and how the standalone locus (flanked by zinc-homeostasis and *polA* genes) is transcriptionally controlled, remains unknown. The genomic neighborhood provides no operonic clue.
5. **Substrate specificity limits.** L-homoserine specificity is inferred from the HSK class rather than measured for this ortholog; possible activity toward homoserine analogs has not been tested.

Proposed Follow-up Experiments / Actions

1. **Recombinant enzyme characterization.** Overexpress and purify PP_0121, and measure steady-state kinetics (K_m for L-homoserine and ATP, k_{cat} , Mg^{2+} dependence, pH optimum). Confirm strict L-homoserine specificity by testing D-homoserine, L-serine, and homoserine analogs.
2. **Structural determination.** Solve the crystal/cryo-EM structure (or validate an AlphaFold model) of ThrB $\pm$ AMP-PNP $\pm$ L-homoserine to confirm the protein-kinase-like fold and the roles of His189/Asp191/Asn196 in metal coordination and catalysis.
3. **Genetic redundancy test in *P. putida*.** Construct single (ΔPP_0121) and double ($\Delta thrB \Delta thrH$) mutants in KT2440 and assay threonine prototrophy on minimal medium to directly confirm the ThrB/ThrH redundancy in this organism.
4. **Site-directed mutagenesis.** Mutate the HxDxxxxN motif residues (e.g., D191A, N196A) and assess loss of kinase activity to validate the predicted catalytic mechanism.
5. **Regulatory/transcriptional mapping.** Use RNA-seq/reporter assays to determine whether *thrB* is monocistronic and whether its expression is modulated by threonine, isoleucine, or the neighboring zinc-homeostasis regulon.
6. **Metabolic flux/complementation.** Test whether *P. putida thrB* complements an *E. coli thrB* mutant, confirming functional interchangeability across the two structural HK families despite the absence of sequence homology.

Supported and Refuted Hypotheses

Supported: - ThrB is a homoserine kinase (EC 2.7.1.39) catalyzing L-homoserine $\rightarrow$ O-phospho-L-homoserine [UniProt;

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 - The enzyme is L-homoserine/ATP specific and operates via an enzyme·ATP·substrate ternary complex [PMID: 25367138, 14759741]. - ThrB is a "type II" homoserine kinase of the protein-kinase-like/APH superfamily, evolutionarily distinct from GHMP ThrB [Pfam/InterPro;

1.

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- ThrB is cytoplasmic (no signal peptide/TM; sequence analysis). - ThrB is functionally redundant with ThrH in *Pseudomonas* [PMID: 1333566, 10220164].

Refuted / not supported: - That single *thrB* deletion causes threonine auxotrophy in *Pseudomonas* (refuted; requires *thrB* + *thrH* double loss [PMID: 1333566, 10220164]). - That *Pseudomonas* ThrB is homologous to the *E. coli* GHMP-family homoserine kinase (refuted by sequence non-similarity and APH domain assignment).

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